

Electronic Design Verification and Validation: Excellent Opportunities

Here, we discuss the scope of a career in this domain, entry-level roles, skills expected and the pay packages, with advice from industry experts

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Sometimes used interchangeably, verification and validation ensure that the system being created will meet the requirements of the customer. When the specifications are validated and the system is verified against the specifications, the concluding system fulfills its purpose. Sounds simple, right? But it is pretty challenging and demanding being a verification and validation engineer.

"The purpose of verification and validation is to ensure 'first time right silicon' or few silicon revisions, as revisions are costly in terms of money, effort and time. To ensure correct functionality, designs need to be validated across various parameters, which involves different types of analysis," says Deep Masiwal, CEO, Adroit IC Design.

From ensuring 'does this thing work right' (verification) aspect to 'you built the right thing' (validation), these engineers ensure designs meet requirements and experience the pain of product reworks, pressure of meeting deadlines for the products to be released as per customers' demands, ensure the verification and validation processes are effective, avoid additional expenses and delays, ensure product quality and reduce prototyping costs.

Scope

There is an excellent scope and opportunity in the field of verification and validation believes Rahul Malvi, CEO, Wafer Space. Talking about verification, he says, "Verification occupies the largest portion of the design cycle and, with the size of designs today, it gets more

and more complex and critical to the success of the semiconductor vendor."

On a similar note, Srinivas Mandavilli, country manager, Imagination Technologies India believes that SoC designs are only to get larger and more complex. He says, "In many cases, verification consumes more than 70 per cent of the total design effort. Opportunities in verification and validation are going to increase over the next few years and get more specialised."

Masiwal says, "The total percentage of verification and validation projects carried out in India is 50-60 per cent, which is a pretty big number and highlights its significance in the industry. One can make a very good career in this field."

"A person who gets into verification and validation has a holistic view of the design at a very early stage of his/her career, and this opens up a wide career path. With this exposure, the person has an option to switch to design phase to become a very good systems architect who understands full systems perspective from functional to error scenarios, which can be taken care in the early design," says D. Srinivasan, managing director, Mobiveil.

It is fundamental to appreciate this fact that, in general, the number of designs that start from scratch has been on a decline for a few years now, across this industry, believes Srinivasan Venkataramanan, chief technology officer, CVC Pvt Ltd. He says, "It is not a bad scene but rather a reflection of market which demands more and more integrated devices such as smartphones and chipsets, and ever-shrinking time-to-market schedules. Hence the term 'designers' in

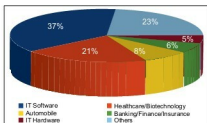


Fig. 1: Top five industries hiring verification/validation professionals (Data courtesy: TimesJobs.com)

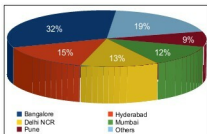


Fig. 2: Top five locations with verification/validation jobs (Data courtesy: TimesJobs.com)

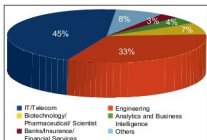


Fig. 3: Top five functional areas hiring verification/validation professionals (Data courtesy: TimesJobs.com)

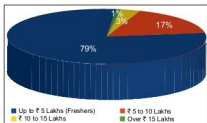


Fig. 4: Salary break-up for verification/validation professionals (Data courtesy: TimesJobs.com)